Supplementary Table S1. A list of Cochrane reviews examining delirium intervention outside of ED since 2012

Study author/year	Main intervention	Setting	Lessons learned for ED	Number of Studies
Woodhouse R et al. 2019 [14]	Software-based identification of high-risk medication	Long term care	Electronic tools and pharmacy led intervention could help to identify high-risk medications	3 cluster randomized controlled trials
Burry L et al. 2018 [15]	Antipsychotic agent	Non-ICU inpatient ward	Antipsychotics did not improve delirium duration or disposition	9 randomized and quasi-randomized trials
Herling SF et al. 2018 [16]	Delirium prevention study (pharmacological)	ICU	Antipsychotics did not prevent delirium.	12 RCT's
Yu A et al. 2018 [17]	Cholinesterase inhibitor	Non-ICU inpatient ward	Insufficient evidence for Cholinesterase inhibitor to prevent delirium	1 RCT
Punjasawadwong Y et al. 2018 [18]	Electroencephalogram	Perioperative	EEG monitoring as an option for delirium assessment	6 RCT's
Siddiqi N et al. 2016 [19]	Delirium prevention study (pharmacological and non-pharmacological treatment)	Mixed setting (None ED)	A strong evidence of multi-component interventions for delirium	39 RCT's
Candy B et al. 2012 [20]	Drug therapy	Delirium with AIDS in the palliative care	Insufficient evidence of drug therapy for delirium symptoms	1 RCT

ICU: Intensive Care Unit, EEG: Electroencephalogram, RCT: Randomized controlled tria

Supplementary Table S2. Ovid MEDLINE Strategy

1. exp Aged/
2. exp Aging/
3. exp Geriatrics/
4. exp Geriatric Nursing/
5. exp Geriatric Assessment/
6. exp Housing for the Elderly/
7. exp Homes for the Aged/
8. exp Nursing Homes/
9. exp Senior Centers/
10. exp "Activities of Daily Living"/
11. ag?ing.tw.
12. elder$.tw.
13. senium.tw.
14. senior citizen$.tw.
15. seniors.tw.
16. geriatric$.tw.
17. psychogeriatric$.tw.
18. gerontology.tw.
19. ((old or older or aged) adj3 (people or person$ or patient$ or adult$ or population$)).tw.
20. postmenopausal wom#n.tw.
21. post menopausal wom#n.tw.
22. frail.tw.
23. frailty.tw.
24. assisted living.tw.
25. nursing home$.tw.
26. community dwelling.tw.
27. mini mental state exam$.tw.
28. mini mental status exam$.tw.
29. mmse.tw.
30. gds.tw.
31. activities of daily living.tw.
32. adl.tw.
33. or/1-32
34. Emergency Medicine/
35. exp Emergency Service, Hospital/
36. Emergency Medical Services/
37. exp Emergency Services, Psychiatric/
38. (emergency adj3 (department$ or ward$ or room$ or service$ or medicine or visit$ or unit$)).tw.
39. ED.tw.
40. ER.tw.
41. A&E.tw.
42. accident & emergency.tw.
43. or/34-42
44. exp Delirium/

45. Confusion/
46. exp Cognitive Dysfunction/
47. exp Postoperative Cognitive Complications/
48. delirium.tw.
49. del#rious.tw.
50. deliria.tw.
51. confusion.tw.
52. confused.tw.
53. disorientation.tw.
54. disoriented.tw.
55. ((neurocognitive or neurocognition or cognitive or cognition) adj3 (impairment$ or declin$ or defect$ or deterioration$ or disorder$ or dysfunction$)).tw.
56. organic brain syndrome$.tw.
57. attention disturbance$.tw.
58. neurobehavioral manifestation$.tw.
59. or/44-58
60. 33 and 43 and 59

Embase Strategy
#69. #39 AND #47 AND #68
#68. #48 OR #49 OR #50 OR #51 OR #52 OR #53 OR #54 OR #55 OR #56 OR #57 OR #58 OR #59 OR #60 OR #61 OR #62 OR #63 OR #64 OR #65 OR #66 OR #67
#67. 'neurobehavioral manifestation*':ti,ab
#66. 'attention disturbance':ti,ab
#65. 'organic brain syndrome':ti,ab
#64. ((neurocognitive OR neurocognition OR cognitive OR cognition) NEAR/3 (impairment* OR declin* OR defect* OR deterioration* OR disorder* OR dysfunction*)):ti,ab
#63. disoriented:ti,ab
#62. disorientation:ti,ab
#61. confused:ti,ab
#60. confusion:ti,ab
#59. delerious:ti,ab
#58. delirious:ti,ab
#57. delirium:ti,ab
#56. 'cognitive defect'/de
#55. 'disorientation'/exp
#54. 'attention disturbance'/exp
#53. 'organic brain syndrome'/exp
#52. 'mild cognitive impairment'/exp
#51. 'postoperative cognitive dysfunction'/exp
#50. 'cognitive defect'/de
#49. 'confusion'/exp
#48. 'delirium'/exp
#47. #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46
#46. 'accident and emergency':ti,ab

#45. er:ti,ab
#44. ed:ti,ab
#43. (emergency NEAR/3 (department$ OR ward$ OR room$ OR service$ OR medicine OR visit$ OR unit$)):ti,ab
#42. 'emergency health service'/exp
#41. 'emergency medicine'/exp
#40. 'emergency ward'/exp
#39. #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38
#38. adl:ti,ab
#37. 'activities of daily living':ti,ab
#36. gds:ti,ab
#35. mmse:ti,ab
#34. 'mini mental state exam*':ti,ab
#33. 'mini mental status exam*':ti,ab
#32. 'community dwelling':ti,ab
#31. 'nursing home$':ti,ab
#30. 'assisted living':ti,ab
#29. frailty:ti,ab
#28. frail:ti,ab
#27. 'post menopausal women':ti,ab
#26. 'post menopausal woman':ti,ab
#25. 'postmenopausal women':ti,ab
#24. 'postmenopausal woman':ti,ab
#23. ((old OR older OR aged) NEAR/3 (people OR person$ OR patient$ OR adult$ OR population$)):ti,ab
#22. gerontology:ti,ab
#21. psychogeriatric*:ti,ab
#20. geriatric*:ti,ab
#19. seniors:ti,ab
#18. 'senior citizen$':ti,ab
#17. senium:ti,ab
#16. elder*:ti,ab
#15. aging:ti,ab
#14. ageing:ti,ab
#13. 'daily life activity'/exp
#12. 'senior center'/exp
#11. 'nursing home'/exp
#10. 'elderly care'/exp
#9. 'geriatric assessment'/exp
#8. 'geriatric patient'/exp
#7. 'geriatrics rehabilitation'/exp
#6. 'geriatrics surgery'/exp
#5. 'geriatrics psychology'/exp

#4. 'geriatrics nursing'/exp
#3. 'geriatrics'/exp
#2. 'aging'/exp
#1. 'aged'/exp

Cochrane CENTRAL Strategy

#1 MeSH descriptor: [Aged] explode all trees
#2 MeSH descriptor: [Aged] explode all trees
#3 MeSH descriptor: [Geriatrics] explode all trees
#4 MeSH descriptor: [Geriatric Nursing] explode all trees
#5 MeSH descriptor: [Geriatric Assessment] explode all trees
#6 MeSH descriptor: [Housing for the Elderly] explode all trees
#7 MeSH descriptor: [Homes for the Aged] explode all trees
#8 MeSH descriptor: [Nursing Homes] explode all trees
#9 MeSH descriptor: [Senior Centers] explode all trees
#10 MeSH descriptor: [Activities of Daily Living] explode all trees
#11 (aging):ti,ab,kw
#12 (elder*):ti,ab,kw
#13 (senium):ti,ab,kw
#14 ("senior citizen*"):ti,ab,kw
#15 (seniors):ti,ab,kw
#16 (geriatric*):ti,ab,kw
#17 (psychogeriatric*):ti,ab,kw
#18 (gerontology):ti,ab,kw
#19 ((old OR older OR aged) NEAR/3 (people OR person* OR patient* OR adult* OR population*)):ti,ab,kw
#20 ("postmenopausal women"):ti,ab,kw
#21 ("post menopausal women"):ti,ab,kw
#22 (frail):ti,ab,kw
#23 (frailty):ti,ab,kw
#24 ("assisted living"):ti,ab,kw
#25 ("nursing home*"):ti,ab,kw
#26 ("community dwelling"):ti,ab,kw
#27 ("mini mental status exam*"):ti,ab,kw
#28 ("mini mental state exam*"):ti,ab,kw
#29 (mmse):ti,ab,kw
#30 (gds):ti,ab,kw
#31 ("activities of daily living"):ti,ab,kw
#32 (adl):ti,ab,kw
#33 {OR #1-#32}
#34 MeSH descriptor: [Emergency Medicine] explode all trees
#35 MeSH descriptor: [Emergency Service, Hospital] explode all trees
#36 MeSH descriptor: [Emergency Medical Services] this term only
#37 MeSH descriptor: [Emergency Services, Psychiatric] explode all trees

#38 ((emergency NEAR/3 (department* OR ward* OR room* OR service* OR medicine OR visit* OR unit*))):ti,ab,kw
#39 (ED):ti,ab,kw
#40 (ER):ti,ab,kw
#41 (A&E):ti,ab,kw
#42 ("accident and emergency"):ti,ab,kw
#43 {OR #34-#42}
#44 MeSH descriptor: [Delirium] explode all trees
#45 MeSH descriptor: [Confusion] this term only
#46 MeSH descriptor: [Cognitive Dysfunction] explode all trees
#47 MeSH descriptor: [Postoperative Cognitive Complications] explode all trees
#48 (delirium):ti,ab,kw
#49 (delirious):ti,ab,kw
#50 (confusion):ti,ab,kw
#51 (confused):ti,ab,kw
#52 (disorientation):ti,ab,kw
#53 (disoriented):ti,ab,kw
#54 (((neurocognitive OR neurocognition OR cognitive OR cognition) NEAR/3 (impairment* OR declin* OR defect* OR deterioration* OR disorder* OR dysfunction*))):ti,ab,kw
#55 ("organic brain syndrome"):ti,ab,kw
#56 ("attention disturbance"):ti,ab,kw
#57 ("neurobehavioral manifestations"):ti,ab,kw
#58 {OR #44-#57}
#59 #33 AND #43 AND #58

CINAHL Strategy

S28 S13 AND S22 AND S27
S27 S23 OR S24 OR S25 OR S26
S26 delirium OR delirious OR deliria OR confusion OR confused OR disorientation OR disoriented OR "organic brain syndrome*" OR "attention disturbance*" OR "neurobehavioral manifestation*"
S25 (neurocognitive OR neurocognition OR cognitive OR cognition) N3 (impairment* or declin* or defect* or deterioration* or disorder* or dysfunction*)
S24 (MH "Mild Cognitive Impairment")
S23 (MH "Confusion+") OR (MH "Delirium")
S22 S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20 OR S21
S21 "accident & emergency"
S20 "accident and emergency"
S19 "er" OR "ed"
S18 (emergency N3 (department* or ward* or room* or service* or visit* or unit*))
S17 (MH "Emergency Services, Psychiatric")
S16 (MH "Emergency Medical Services")
S15 (MH "Emergency Service+")
S14 (MH "Emergency Medicine")
S13 S1 OR S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12
S12 "assisted living" OR "nursing home*" OR "community dwelling" OR "mini mental state exam*" OR "mini mental status exam*" OR mmse OR gds OR "activities of daily living" OR adl

S11 ((old or older or aged) N3 (people or person* or patient* or adult* or population*))
S10 aging OR ageing OR elder* OR senium OR "senior citizen*" OR seniors OR geriatric* OR psychogeriatric* OR gerontology OR "postmenopausal women" OR "postmenopausal woman" OR "post menopausal women" OR "post menopausal woman" OR frail OR frailty
S9 (MH "Nursing Homes+")
S8 (MH "Housing for the Elderly")
S7 (MH "Geriatric Psychiatry")
S6 (MH "Gerontologic Care")
S5 (MH "Geriatric Assessment+")
S4 (MH "Gerontologic Nursing+")
S3 (MH "Geriatrics")
S2 (MH "Aging+")
S1 (MH "Aged+")

PsycINFO Strategy

S24 S22 NOT S23
S23 PT book
S22 S8 AND S15 AND S21
S21 S16 OR S17 OR S18 OR S19 OR S20
S20 delirium OR delirious OR deliria OR confusion OR confused OR disorientation OR disoriented OR "organic brain syndrome*" OR "attention disturbance*" OR "neurobehavioral manifestation*"
S19 (neurocognitive OR neurocognition OR cognitive OR cognition) N3 (impairment* or declin* or defect* or deterioration* or disorder* or dysfunction*)
S18 DE "Cognitive Impairment"
S17 DE "Mental Confusion"
S16 DE "Delirium"
S15 S9 OR S10 OR S11 OR S12 OR S13 OR S14
S14 "accident & emergency"
S13 "accident and emergency"
S12 "er" OR "ed"
S11 (emergency N3 (department* or ward* or room* or service* or visit* or unit*))
S10 DE "Emergency Services"
S9 DE "Emergency Medicine"
S8 S1 OR S2 OR S3 OR S4 OR S5 OR S6 OR S7
S7 "assisted living" OR "nursing home*" OR "community dwelling" OR "mini mental state exam*" OR "mini mental status exam*" OR mmse OR gds OR "activities of daily living" OR adl
S6 ((old or older or aged) N3 (people or person* or patient* or adult* or population*))
S5 aging OR ageing OR elder* OR senium OR "senior citizen*" OR seniors OR geriatric* OR psychogeriatric* OR gerontology OR "postmenopausal women" OR "postmenopausal woman" OR "post menopausal women" OR "post menopausal woman" OR frail OR frailty
S4 DE "Activities of Daily Living"
S3 DE "Nursing Homes" OR DE "Nursing Home Residents"
S2 DE "Geriatrics" OR DE "Geriatric Assessment" OR DE "Geriatric Psychiatry"
S1 DE "Aging" OR DE "Aging in Place" OR DE "Cognitive Aging" OR DE "Healthy Aging" OR DE "Physiological Aging"

ProQuest Dissertations & Theses Global Strategy

((mainsubject.Exact("aging") OR mainsubject.Exact("geriatrics") OR mainsubject.Exact("nursing homes") OR mainsubject.Exact("activities of daily living")) OR (ti(aging OR elder OR elders OR elderly) OR ab(aging OR elder OR elders OR elderly)) OR (ti(senium OR "senior citizen" OR "senior citizens" OR seniors OR geriatric OR geriatrics OR psychogeriatric OR psychogeriatrics OR gerontology OR "postmenopausal women" OR "postmenopausal woman" OR "post menopausal women" OR "post menopausal woman" OR frail OR frailty) OR ab(senium OR "senior citizen" OR "senior citizens" OR seniors OR geriatric OR geriatrics OR psychogeriatric OR psychogeriatrics OR gerontology OR "postmenopausal women" OR "postmenopausal woman" OR "post menopausal women" OR "post menopausal woman" OR frail OR frailty)) OR (ti("assisted living" OR ("nursing home" OR "nursing homes") OR "community dwelling" OR "mini mental state exam*" OR "mini mental status exam*" OR mmse OR gds OR "activities of daily living" OR adl) OR ab("assisted living" OR ("nursing home" OR "nursing homes") OR "community dwelling" OR "mini mental state exam*" OR "mini mental status exam*" OR mmse OR gds OR "activities of daily living" OR adl)) OR ((old OR older OR aged) N3 (people OR person OR persons OR patient OR patients OR adult OR adults OR population OR populations))) AND (mainsubject.Exact("delirium") OR ((neurocognitive OR neurocognition OR cognitive OR cognition) N3 (impairment OR impairments OR decline OR declines OR declining OR declined OR defect OR defects OR deterioration OR disorder OR disorders OR dysfunction OR dysfunctions OR dysfunctional)) OR (ti(delirium OR delirious OR deliria OR confusion OR confused OR disorientation OR disoriented OR "organic brain syndrome" OR "attention disturbance" OR "attention disturbances" OR "neurobehavioral manifestation" OR "neurobehavioral manifestations") OR ab(delirium OR delirious OR deliria OR confusion OR confused OR disorientation OR disoriented OR "organic brain syndrome" OR "attention disturbance" OR "attention disturbances" OR "neurobehavioral manifestation" OR "neurobehavioral manifestations"))) AND (ti("emergency department*" OR "emergency ward*" OR "emergency room*" OR "emergency service*" OR "emergency medical service*" OR ER OR ED OR "accident and emergency") OR ab("emergency department*" OR "emergency ward*" OR "emergency room*" OR "emergency service*" OR "emergency medical service*" OR ER OR ED OR "accident and emergency"))

Supplementary Table S3. Summary of selected articles.

Author/ Study year	Study Type	Study period	Inclusion/Exclusion (specify how high-risk group chosen for this)	Sample Size	Age (Mean, SD or median, IQR)	% Female	Ethnicity	Intervention	Population	Setting	Country
Al-Aama, T/2010[43]	RCT	October 29, 2007 – February 29, 2008.	Patient at least age 65 years old and admitted to into the hospital through the ED were included. Patients were excluded if they had life expectancy less than 48 hr, unable to speak English, unable to take oral medication, had intracranial bleed or seizures.	145	Placebo: 84.6 (6.2) Melatonin: 84.3 (5.9)	57.4	NA	Melatonin (0.5 mg) vs Placebo	General	ED and inpatient Single center	Canada
Bjorkelund, K/2010[45]	Quasi- Experimental (Prospective measurement before an intervention program and after in a clinical setting; Consent obtained from patients)	April 2003 – April 2004	All patients at least age 65 with suspected hip fracture were screened for cognitive impairment and delirium using the Short Portable Mental Status Questionnaire and Organic Brain Syndrome	275	Control: 82 (7.6) Intervention: 81.1 (7.5)	70.34	NA	Multi-component (O2, IV fluid for nutrition, inc. vital monitoring, pain relief, avoid delay of tests, daily screen for delirium, avoid polypharmacy, and monitoring during perioperative	Older adults with hip fracture	ED and inpatient Single center	Sweden

			Scale, respectively. **Patients with cognitive impairment, severe neuropsychiatric illness, difficulties to communicate and multi-system trauma were excluded.**					**anesthetic period.)**			
Herrman, N/2020 [40]	**Before and after study, retrospective chart review**	**NA**	**Patients with hip fracture one year prior to and one year after guideline implementation were included.**	**308**	**NA**	**NA**	**NA**	**FNB done by all providers trained to perform FNB guided by US.**	**Hip fracture**	**ED**	**US**
LeBlanc, P/2016[37]	**Case-control**	**December 2013 – April 2015**	**Patients at least 60 years old admitted to the ED with a hip fracture and surgically repaired were included.** **Patients with delirium upon arrival or other cognitive status and inability to speak English were excluded.**	**Not reported**	**NA**	**NA**	**NA**	**Femoral Nerve Block (FNB) with US guidance done by trained emergency doctors**	**Hip fracture**	**ED**	**Canada**
Mok, W/2017[38]	**Quality improvement**	**January 2015 – October 2016**	**Patients that underwent hip surgery from January 2015 – October 2016 were included.**	**494**	**NA**	**NA**	**NA**	**Multi-component (CAM, O2, fluid and electrolyte balance, continence management, pain treatment, nutrition, early**	**Hip fracture**	**ED and inpatient unit**	**Singapore**

								mobilization and rehab)			
Naughton, B/2005 [44]	Quality improvement	September 1998 – September 2000	Patients at least age 75 admitted from the ED and hospitalized for at least 4 days were included.	374	Baseline: 81 (6.2) 4 Months: 81 (6.1) 9 Months: 82 (5.9)	63	NA	Multi-component (guideline based medication management, family support, and treatment of medical factors)	General	ED and geriatric unit Single center	US
Noel, C/2019 [46]	Case- Control	March 2012 – November 2014	All patients at 65 years old admitted to the hospital through the ED for screened for delirium using bCAM. Patients discharged from the ED, non-English speaking, nonverbal, deaf and patients with incomplete data were excluded.	228	Nondelirious ED Patients: 73 (69-80) Delirious ED Patients: 75 (68-83)	55.3	Nonwhite 9.8%	Foley catheter, ED medication use, and ED LOS.	Older adults	ED	US
Schrijver, E/2018 [42]	RCT	November 5, 2012 – March 1, 2015	Evaluated patients with a delirium risk assessment tool value of safety management system (VMS). Score of 1 or greater out of 3 are considered increased risk and included. Patients who required immediate	245	83.5 (6.3)	55.4	NA	Haloperidol (1 mg) vs Placebo	General	ED and inpatient Six institutions	Netherlands

			surgery, had suspected delirium, other cognitive illnesses, and communication barriers were excluded.								
Sun, H/2019[39]	Before and after study	NA	Patients at least 65 years old who arrives at the ED or admitted to the ICU or medical floor and determined by the charge nurse to be at high risk for delirium were included.	Not reported	NA	NA	NA	Non-pharmacological (puzzles, music, re-orienting patients and advocating for patients.)	Older adults	ED, inpatient unit, and ICU	US
Yip, D/2019[41]	Quality improvement	January 1, 2017 – December 31, 2017.	Hip fracture patients from January 1, 2017 – December 31, 2017, were included.	282	NA	NA	NA	Fascia illiaca block (FIB) done by EM physician or anesthesiology consultation	Hip fracture	ED	US